Exercises: Differentiation in Several Variables

Mathematical Foundations for Machine Learning

Partial Derivatives

Exercise 1.1 [Easy]
Compute all partial derivatives of $f(x, y) = x^3y + 2xy^2 - 5y$.

Exercise 1.2 [Easy]
Let $f(x, y, z) = e^{xz}\sin(y)$. Compute $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$, and $\frac{\partial f}{\partial z}$.

Exercise 1.3 [Medium]
Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} \dfrac{x^2y}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

the derivative is undefined because if you take lim h-> 0 (h, h) and lim h->0 (h,-h) its different limits, so depends on from where you approach it is different limits.

Compute $\frac{\partial f}{\partial x}(0, 0)$ and $\frac{\partial f}{\partial y}(0, 0)$ using the limit definition.

Exercise 1.4 [Medium]
Let $f(x, y) = \ln(x^2 + y^2)$ for $(x, y) \neq (0, 0)$. Use the quotient rule to compute both partial derivatives and simplify.

Gradients and Jacobians

Exercise 2.1 [Easy]
Compute the gradient of $f(x, y, z) = x^2y + yz^3 - 3xz$.

Exercise 2.2 [Easy]
Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by $f(x, y) = \begin{pmatrix} x^2 + y \\ xy \end{pmatrix}$. Compute the Jacobian $Df(x, y)$.

Exercise 2.3 [Medium]
Let $f : \mathbb{R}^3 \to \mathbb{R}^2$ be defined by $f(x, y, z) = \begin{pmatrix} e^{x+y} \\ \sin(xz) \end{pmatrix}$. Compute the Jacobian $Df(x, y, z)$ and verify that each row is the transpose of the gradient of the corresponding component.

Exercise 2.4 [Medium]
Let $f : \mathbb{R}^n \to \mathbb{R}$ be defined by $f(x) = \langle a, x\rangle + b$ where $a \in \mathbb{R}^n$ and $b \in \mathbb{R}$. Show that $\nabla f(x) = a$ for all x. What is the Jacobian?

Total Differentiability

Exercise 3.1 [Easy]
Let $f(x, y) = 3x + 2y - 1$. Show that f is totally differentiable at every point and identify the matrix A from the definition of total differentiability.

To be totally differentiable we need all the partials to be defined + R(h) vanish before ||h|| -> else, it could happ en like the example where the derivative was not defined but the partials existed.

Exercise 3.2 [Medium]
Consider the function

$$f(x, y) = \begin{cases} \dfrac{xy}{\sqrt{x^2 + y^2}} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

Show that both partial derivatives exist at the origin, but f is **not** totally differentiable at $(0, 0)$.

Exercise 3.3 [Hard]
Prove: if $f : \mathbb{R}^n \to \mathbb{R}^m$ is totally differentiable at x, then f is continuous at x.

Hint: Use the definition $f(x + h) = f(x) + Ah + R(h)$ and the fact that $\|R(h)\| \leq \|h\|$ for $\|h\|$ small enough.

Chain Rule and Product Rule

Exercise 4.1 [Easy]
Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ with $f(x, y) = (x + y,\, xy)$ and $g : \mathbb{R}^2 \to \mathbb{R}$ with $g(u, v) = u^2 + v$. Compute $D(g \circ f)(x, y)$ using the chain rule.

Exercise 4.2 [Easy]
Let $f(x, y) = x^2$ and $g(x, y) = e^y$. Use the product rule to compute $\nabla(f \cdot g)(x, y)$.

Exercise 4.3 [Medium]
Let $\gamma(t) = (\cos t,\, \sin t)$ be a curve in $\mathbb{R}^2$ and $f(x, y) = x^2 + y^2$. Using the chain rule, compute $\frac{d}{dt} f(\gamma(t))$. Interpret your result geometrically.

Exercise 4.4 [Hard]
Let $f : \mathbb{R}^n \to \mathbb{R}^m$ and $g : \mathbb{R}^m \to \mathbb{R}^l$ be totally differentiable. Using the definition of total differentiability, prove the chain rule:

$$D(g \circ f)(x) = Dg(f(x))\, Df(x).$$

Hessian and Schwarz's Theorem

Exercise 5.1 [Easy]
Compute the Hessian matrix $H_f(x, y)$ of $f(x, y) = x^3 + 3x^2y - y^3$.

Exercise 5.2 [Easy]
Compute the Laplacian Δf of $f(x, y, z) = x^2 + y^2 + z^2$.

Exercise 5.3 [Medium]
Let $f(x, y) = e^{xy}$. Compute the full Hessian H_f and verify Schwarz's theorem by checking $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$.

Exercise 5.4 [Hard]
Let $f : \mathbb{R}^n \to \mathbb{R}$ be given by $f(x) = \frac{1}{2} x^T A x$ where $A \in \mathbb{R}^{n \times n}$ is a constant matrix. Show that $H_f(x) = \frac{1}{2}(A + A^T)$ for all x. What happens if A is symmetric?

Directional Derivatives and Steepest Ascent

Exercise 6.1 [Easy]
Let $f(x, y) = x^2 - xy + y^2$. Compute the directional derivative of f at the point $(1, 2)$ in the direction $v = \frac{1}{\sqrt{2}}(1, 1)^T$.

Exercise 6.2 [Medium]
Let $f(x, y) = e^{-(x^2+4y^2)}$. At the point $(1, 0)$:

(a) Find the direction of steepest ascent (using the Euclidean norm).

(b) Find the maximum rate of increase.

Exercise 6.3 [Medium]
Let $f(x, y) = x^2 + y^2$ and suppose we measure the direction v in the ℓ_∞ norm, i.e., $\|v\|_\infty = 1$. Find the direction of maximal ascent at the point $(3, 1)$.
Hint: The direction of maximal ascent under $\|v\|_\infty = 1$ is $v_i = \text{sign}((\nabla f)_i)$.

Exercise 6.4 [Hard]
Prove that among all unit vectors v with $\|v\|_2 = 1$, the directional derivative $D_v f(x)$ is maximized when $v = \frac{\nabla f(x)}{\|\nabla f(x)\|_2}$ (assuming $\nabla f(x) \neq 0$), and that the maximum value is $\|\nabla f(x)\|_2$.
Hint: Use the Cauchy–Schwarz inequality.

Level Sets and Gradient Orthogonality

Exercise 7.1 [Easy]
Let $f(x, y) = x^2 + 4y^2$. Sketch (or describe) the level set $L_f(4) = \{(x, y) : f(x, y) = 4\}$. What is ∇f at the point $(2, 0)$? Verify that it is orthogonal to the level curve at that point.

Exercise 7.2 [Medium]
Let $f \in C^1(\mathbb{R}^n)$ and let $\gamma : [0, 1] \to \mathbb{R}^n$ be a curve contained in the level set $L_f(c) = \{x : f(x) = c\}$. Prove that $\nabla f(\gamma(t))$ is orthogonal to $\dot{\gamma}(t)$ for all $t \in [0, 1]$.

Mean Value Theorem and Lipschitz Continuity

Exercise 8.1 [Easy]
Let $f(x, y) = x^2 + y^2$ defined on $\Omega = \{(x, y) : x^2 + y^2 \leq 1\}$ (the closed unit disk). Find an upper bound for $|f(x) - f(y)|$ for $x, y \in \Omega$ using the gradient bound.

Hint: Bound $\|\nabla f(z)\|_2$ on Ω.

Exercise 8.2 [Medium]
Let $f : \mathbb{R}^n \to \mathbb{R}$ be in $C^1(\mathbb{R}^n)$ and suppose $\|\nabla f(x)\|_2 \leq L$ for all $x \in \mathbb{R}^n$. Using the fundamental theorem of calculus along the straight-line path $\gamma(t) = x + t(y - x)$, prove that $|f(y) - f(x)| \leq L\|y - x\|_2$.

Exercise 8.3 [Hard]
Let $f : \mathbb{R}^n \to \mathbb{R}$ be in $C^1(\mathbb{R}^n)$. Suppose f is L-Lipschitz continuous: $|f(x) - f(y)| \leq L\|x - y\|_2$ for all x, y. Prove that $\|\nabla f(x)\|_2 \leq L$ for all x.

Hint: Consider $y = x + tv$ for unit vector v and take $t \to 0$.

Mixed Challenging Exercises

Exercise 9.1 [Hard]
Let $f : \mathbb{R}^n \to \mathbb{R}$ be twice continuously differentiable. Show that the Hessian $H_f(x)$ satisfies:

$$H_f(x) = \lim_{h \to 0} \frac{f(x + h) - f(x) - \langle \nabla f(x), h \rangle - \frac{1}{2} h^T H_f(x) h}{\|h\|^2} = 0$$

by writing the second-order Taylor expansion of f around x.

Exercise 9.2 [Hard]
Let $f : \mathbb{R}^n \to \mathbb{R}$ be defined by $f(x) = \|Ax - b\|_2^2$ where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. Compute $\nabla f(x)$ and $H_f(x)$. Use these to find the critical points of f.

Exercise 9.3 [Hard]

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be given by $f(x, y) = e^{-\langle z, Bz \rangle}$ where $z = (x, y)^T$ and $B = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$.

(a) Compute $\nabla f(x, y)$ using the chain rule.

(b) Show that ∇f is orthogonal to the level sets of f.

(c) Is f Lipschitz continuous on $\mathbb{R}^2$? Justify your answer.